

Intermediate Macroeconomics: Lecture 5 (Ch. 4)

ECON 311 – Intermediate Macroeconomics

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Firm's Problem

- Firms decide how many workers and machines to use in production

$$Y = F(K, L)$$

- Choose K and L to maximize Profits = Revenue - Cost

- Profit is defined as total revenue minus total costs:

$$\max_{K,L} \pi = F(K, L) - rK - wL$$

where π is the profit, r is the rental rate of capital, and w is the wage rate

- The term rK is the price per unit of capital multiplied by the number of capital inputs used (total cost of all capital)
- wL is the price of labour multiplied by the number of labour inputs used (total cost of all labour)

Firm's Problem

- Assuming a perfectly competitive market
- Firms take rental rate r and wage rate w as given
 - No firm has control to change input prices
 - No matter how much or how little L or K a firm uses, the prices won't change
- Input prices will change only as the result of an exogenous force in the market for K or L
- For simplicity, the price of the output is normalized to one

Firm's Problem

- The rental rate and wage rate are taken as given under perfect competition
 - Hire capital until $MPK = r$
 - Hire labour until $MPL = w$
 - If $MPL > w$, the amount of output produced by an additional worker is greater than w
 - Profits increase, this will continue until the cost of hiring a worker is equal to the amount of extra output the worker produces
 - The same reasoning holds for capital

Marginal Products

- Using the Codd-Douglas production function: $Y = AK^{1/3}L^{2/3}$
- The marginal product of labour (MPL) is the additional output that is produced when one unit of labour is added, holding all other inputs constant:

$$MPL = \frac{2}{3}A \left(\frac{K}{L}\right)^{1/3} = \left(\frac{2}{3}\right) \left(\frac{Y}{L}\right)$$

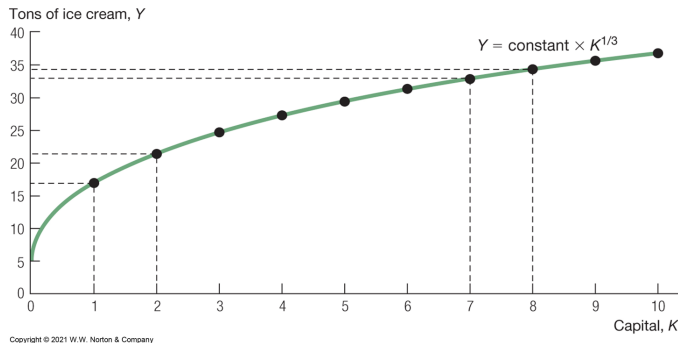
- The marginal product of capital (MPK) is the additional output that is produced when one unit of capital is added, holding all other inputs constant:

$$MPK = \frac{1}{3}A \left(\frac{L}{K}\right)^{2/3} = \left(\frac{1}{3}\right) \left(\frac{Y}{K}\right)$$

- In a Cobb-Douglas production function, the marginal product of an input is equal to the product of the factor's exponent times the average amount that each unit of the factor produces

Diminishing Marginal Product of Capital in Production

- When labour input is held constant, Each additional unit of K increases output by less and less



Diminishing Returns

- Formally,

$$\frac{\partial^2 F}{\partial K^2} < 0 \text{ and } \frac{\partial^2 F}{\partial L^2} < 0$$

- For example, suppose we have five computers
 - The first five people we hire are very productive
 - But as we add more and more workers, there is less for them to do with only five computers among them

Solving the Model: General Equilibrium

- In general equilibrium, we solve for not only one single market but both market of labour and capital
- Assume Number of perfectly competitive firms = 1, we could have more firms if we wished and nothing would change since our model features constant returns to scale
- Demand for capital (labour) by our representative firm = aggregate quantity of capital (labour) in the economy

Solving the Model: General Equilibrium

- Five endogenous variables
 - Output (Y)
 - The amount of capital (K)
 - The amount of labour (L)
 - The wage (w)
 - The rental price of capital (r)

Solving the Model: General Equilibrium

- Five equations

- The production function

$$Y = AK^{1/3}L^{2/3}$$

- The rule for hiring capital

$$\frac{1}{3} \frac{Y}{K} = r$$

- The rule for hiring labour

$$\frac{2}{3} \frac{Y}{L} = w$$

- Supply equals the demand for labour

$$L = \bar{L}$$

- Supply equals the demand for capital

$$K = \bar{K}$$

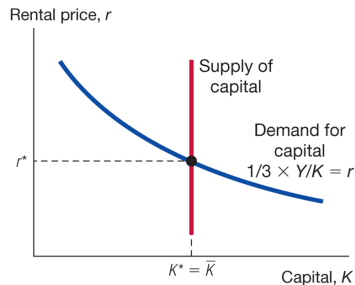
- The supply of labour and capital are assumed to be perfectly inelastic, the economy will supply $\bar{L}$ amount of labour and $\bar{K}$ amount of capital regardless of the price

Solving the Model: Solution

- A solution to the model
 - A new set of equations that express the five unknowns in terms of the parameters and exogenous variables
 - There should be five of them as we have five endogenous variables
 - Endogenous variables are on the left side, and the parameters and exogenous variables that describe market behavior in equilibrium are on the right side.
- General equilibrium
 - Solution to the model when more than a single market clears

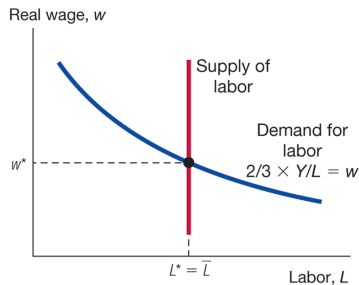
Supply and Demand in the Capital and labour Markets

- The solution easiest to see graphically (supply and demand for K and L)
- The supplies of K and L exogenously given (vertical lines)
- The demand curves for capital and labour are based on the hiring rules
 - They trace out exactly the marginal product schedules for K and L



(a) The capital market

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(b) The labor market

- The solution implies that
 - Firms employ all the supplied capital and labour in the economy
 - The production function is evaluated with the given supply of inputs:

$$Y^* = F(\bar{K}, \bar{L}) = \bar{A}\bar{K}^\alpha \bar{L}^{1-\alpha}$$

- $w = MPL$ evaluated at the equilibrium values of Y , K , and L
- $r = MPK$ evaluated at the equilibrium values of Y , K , and L

Interpreting the Solution

- The equilibrium wage is proportional to output per worker: Output per worker = (Y/L)
- The equilibrium rental rate is proportional to output per capital: Output per capital = (Y/K)
- In the United States, empirical evidence shows:

$$\frac{w^* L^*}{Y^*} = \frac{2}{3} \text{ and } \frac{r^* K^*}{Y^*} = \frac{1}{3}$$

- The factor shares of the payments are equal to the exponents on the inputs in the Cobb-Douglas function

All income is paid to capital or labour

- Verifies the assumption of perfect competition
- One of the implications of perfect competition is zero profits

$$\pi = Y^* - r^*K^* - w^*L^* = 0$$

- Also verifies that production equals spending equals income

$$w^*L^* + r^*K^* = Y^*$$

Income = production

- This is one of the fundamental relationships of national income accounting.

Analyzing the Production Model

Development accounting:

- The use of a model to explain differences in incomes across countries

$$y^* = \bar{A}\bar{k}^{1/3}$$

- For now, setting the productivity parameter = 1

$$y^* = \bar{k}^{1/3}$$

The Empirical Fit of the Production Function

- By using different countries' GDP per capita and capital per capita, we can check the empirical fit of the production function
- If the productivity parameter is 1, the model overpredicts GDP per capita
 - Assuming that the productivity parameter is constant across countries, and does not vary over time
- Capital per person varies tremendously across countries ranging from a low in Burundi of less than 1 percent of the U.S. value to a high in Switzerland of about 30 percent greater

The Model's Prediction for Per Capita GDP (United States = 1)

- The predicted value is simply $y = (0.93)^{1/3} = 0.976$ for Japan

TABLE 4.3

The Model's Prediction for Per Capita GDP (U.S. = 1)

Country	Observed capital per person, $\bar{k}$	Predicted per capita GDP $y = \bar{k}^{1/3}$	Observed per capita GDP
U.S.	1.000	1.000	1.000
Switzerland	1.309	1.094	1.151
Japan	0.930	0.976	0.734
U.K.	0.693	0.885	0.714
Italy	0.804	0.930	0.680
Spain	0.731	0.901	0.640
China	0.276	0.651	0.279
Brazil	0.202	0.587	0.252
South Africa	0.174	0.559	0.214
India	0.081	0.433	0.117
Burundi	0.005	0.173	0.015

Predicted per capita GDP is computed as $\bar{k}^{1/3}$, that is, assuming no differences in productivity across countries. Data correspond to the year 2017 and are divided by the values for the United States.

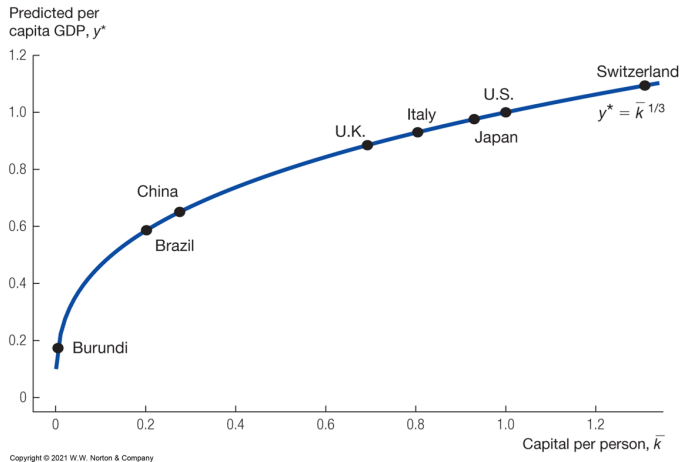
Source: Penn World Tables, Version 9.1.

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The Empirical Fit of the Production Function

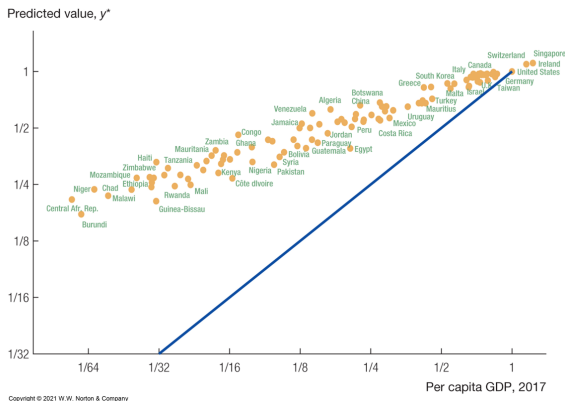
- With no difference in productivity ($\bar{A} = 1$), the model predicts smaller differences in income across countries than we observe
- Diminishing returns to capital implies that
 - countries with low K will have a high MPK
 - countries with a lot of K will have a low MPK, and cannot raise GDP per capita by much through more capital accumulation
- Differences in capital per person translate into much smaller differences in GDP because of diminishing MPK
- Therefore, gaps in income between rich and poor countries are predicted to be smaller than they are

Predicted Per Capita GDP in the Production Model



The Model's Prediction for Per Capita GDP

- If the model were successful in explaining incomes, countries should lie close to the solid 45-degree line
- Instead, the model predicts that most countries should be substantially richer than they are



Productivity Differences: Improving the Fit of the Model

- The productivity parameter measures how efficiently countries are using their factor inputs – Total factor productivity (TFP)
 - A lower level of TFP implies that workers produce less output for any given level of capital per person
- If TFP is not equal to one for every countries, it would lead to a better model
- But, we have data limitation in that there is no real direct measure of TFP
- We have to exploit the fact that we have data on every term in the equation other than TFP
 - That's why TFP is sometime refer as Solow's residual
- Therefore, we can calculate the level of TFP for each country that would be needed to make the equation hold exactly

Measuring TFP so the Model Fits Exactly

- In order for our model to match the data, poor countries must be very inefficient in production, or they must have low TFP
- For example, Italy = $\frac{y}{k^{1/3}} = 0.68/0.93 = 0.731$

Measuring TFP So the Model Fits Exactly

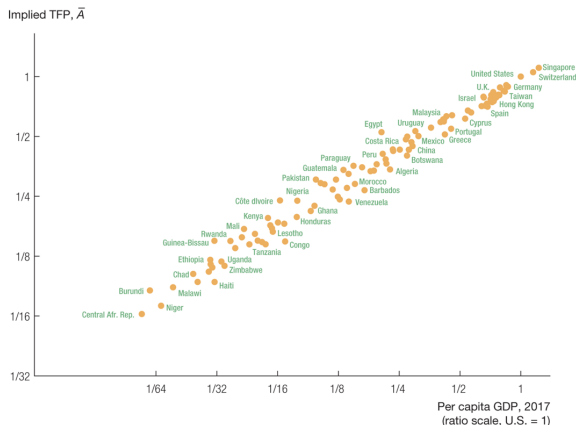
Country	Per capita GDP (y)	$\bar{k}^{1/3}$	Implied TFP ($\bar{A}$)
United States	1.000	1.000	1.000
Switzerland	1.151	1.094	1.052
U.K.	0.714	0.885	0.807
Japan	0.734	0.976	0.752
Italy	0.680	0.930	0.731
Spain	0.640	0.901	0.710
China	0.279	0.651	0.429
Brazil	0.252	0.587	0.429
South Africa	0.214	0.559	0.383
India	0.117	0.433	0.270
Burundi	0.015	0.173	0.084

Calculations are based on the equation $y = \bar{A}\bar{k}^{1/3}$. Implied productivity $\bar{A}$ is calculated from data on y and $\bar{k}$ for the year 2010, so that this equation holds exactly as $\bar{A} = y/\bar{k}^{1/3}$.

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Measuring TFP so the Model Fits Exactly

- Notice that TFP and per capita GDP are highly correlated
- Also, TFP varies substantially across countries
- This is the motivation for examining economic growth and understanding differences between rich and poor countries



Understanding TFP Differences

Based on the simple production function:

- Output differences between the richest and poorest countries?
 - Differences in capital per person explain about one-third of the difference
 - TFP explains the remaining two-thirds
- Thus, rich countries are rich because
 - they have more capital per person
 - more importantly, they use labour and capital more efficiently
- Why are some countries more efficient at using capital and labour?

Understanding TFP Differences - Human Capital

- Human Capital: the stock of knowledge and skills that make people more productive members of society
 - e.g., Education, training, health, etc
- A lot of time researchers measure it through *returns to education*: the value of the increase in wages from additional schooling
- There are significant returns to education within a country
 - In the United States each year of education seems to increase future wages by 7 percent
 - A four-year education may raise wages by about 28 percent (over entire lifetime)
 - In developing countries, returns can be even higher—up to 10 percent or even 13 percent per year
 - The typical student learning the basic skills associated with literacy and arithmetic may have higher returns than a college education.
- Accounting for human capital can reduce the residual

Understanding TFP Differences - Technology

- Richer countries may use more modern and efficient technologies than poor countries
 - Increases productivity parameter
- But, how does the technology diffuse across countries?

Understanding TFP Differences - Institutions

- Well-defined institutions and laws create a climate for economic growth
- Institutions are in place to foster human capital and technological growth
 - Property rights
 - The rule of law
 - Government systems
 - Contract enforcement
- At the time of the collapse of the Berlin Wall in 1989, the standards of living in East and West Germany were substantially different
- Difference between Civil law and Common law can lead to difference in economic development

Understanding TFP Differences - Misallocation

- Misallocation
 - Resources not being put to their best use
- Examples
 - Inefficiency of state-run resources
 - Political interference
- Inefficient land policy: small land size $\rightarrow$ farm cannot be mechanize

Evaluating the Production Model

- Per capita GDP is higher if capital per person is higher and if factors are used more efficiently.
- Constant returns to scale imply that output per person can be written as a function of capital per person
- Capital per person is subject to strong diminishing returns because the exponent is much less than one

Weaknesses of the Model

- The production model has partial empirical success
- In the absence of TFP, the production model predicts only a small differences in income
- The model provided little explanation for why countries exhibit different TFP levels and different capital
- Why do economies grow over time?
- This is what we explore in the next two chapters

Next Lecture

- Next lecture will start Ch.5